

OBJECT ORIENTED PROGRAMMING WITH C++

DSA0103

CO3-AT2-LABORATORY PERFORMANCE

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1.write a c++ program to create a class for a student with a constructor and a destructor

Code:

```
#include <iostream>

using namespace std;

class Student {
    string name;
    int age;
public:
    Student(string n, int a) {
        name = n;
        age = a;
    }
    void display() {
        cout << "Name: " << name << endl;
        cout << "Age: " << age << endl;
    }
    ~Student() {
        cout << "Student object destroyed";
    }
};

int main() {
    Student s("Ravi", 20);
    s.display();
}
```

Output:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class Student { 5 string name; 6 int age; 7 8 public: 9 Student(string n, int a) { 10 name = n; 11 age = a; 12 } 13 14 void display() { 15 cout << "Name: " << name << endl; 16 cout << "Age: " << age << endl; 17 } 18 19 ~Student() { 20 cout << "Student object destroyed"; 21 } 22 }; 23 24 int main() { 25 Student s("Ravi", 20); 26 s.display(); 27 }</pre>		<pre>Name: Ravi Age: 20 Student object destroyed === Code Execution Successful ===</pre>

2.write a c++ program to create a class for a book with a constructor and a destructor

Code:

```
#include <iostream>

using namespace std;

class Book {
    string title;
    float price;
public:
    Book(string t, float p) {
        title = t;
        price = p;
    }

    void display() {
        cout << "Title: " << title << endl;
        cout << "Price: " << price << endl;
    }

    ~Book() {
        cout << "Book object destroyed";
    }
}
```

```
};
```

```
int main() {  
    Book b("C++ Programming", 450);  
    b.display();  
}
```

Output:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class Book { 5 string title; 6 float price; 7 8 public: 9 Book(string t, float p) { 10 title = t; 11 price = p; 12 } 13 14 void display() { 15 cout << "Title: " << title << endl; 16 cout << "Price: " << price << endl; 17 } 18 19 ~Book() { 20 cout << "Book object destroyed"; 21 } 22 }; 23 24 int main() { 25 Book b("C++ Programming", 450); 26 b.display(); 27 }</pre>		<pre>Title: C++ Programming Price: 450 Book object destroyed === Code Execution Successful ===</pre>

3.write a c++ program to create a class for a rectangle with a constructor and a destructor

Code:

```
#include <iostream>  
  
using namespace std;  
  
class Rectangle {  
    int length, width  
  
public:  
    Rectangle(int l, int w) {  
        length = l;  
        width = w;  
    }  
  
    void area() {
```

```

        cout << "Area: " << length * width << endl;
    }
    ~Rectangle() {
        cout << "Rectangle object destroyed";
    }
};

int main() {
    Rectangle r(10, 5);

    r.area();
}

```

Output:

The screenshot shows a C++ IDE with a file named 'main.cpp'. The code is as follows:

```

1 #include <iostream>
2 using namespace std;
3
4 class Rectangle {
5     int length, width;
6
7 public:
8     Rectangle(int l, int w) {
9         length = l;
10        width = w;
11    }
12
13    void area() {
14        cout << "Area: " << length * width << endl;
15    }
16
17    ~Rectangle() {
18        cout << "Rectangle object destroyed";
19    }
20 };
21
22 int main() {
23     Rectangle r(10, 5);
24     r.area();
25 }

```

The IDE has a 'Run' button. The output window on the right shows the following text:

```

Area: 50
Rectangle object destroyed

=== Code Execution Successful ===

```

4.write a c++ program to create a class for a car with a constructor and a destructor

Code:

```

#include <iostream>

using namespace std;

class Car {
    string brand;
    int year;
}

```

```

public:
    Car(string b, int y) {
        brand = b;
        year = y;
    }
    void display() {
        cout << "Brand: " << brand << endl;
        cout << "Year: " << year << endl;
    }
    ~Car() {
        cout << "Car object destroyed";
    }
};

int main() {
    Car c("Toyota", 2024);
    c.display();
}

```

Output:

main.cpp	Run	Output
<pre> 1 #include <iostream> 2 using namespace std; 3 4 class Car { 5 string brand; 6 int year; 7 8 public: 9 Car(string b, int y) { 10 brand = b; 11 year = y; 12 } 13 14 void display() { 15 cout << "Brand: " << brand << endl; 16 cout << "Year: " << year << endl; 17 } 18 19 ~Car() { 20 cout << "Car object destroyed"; 21 } 22 }; 23 24 int main() { 25 Car c("Toyota", 2024); 26 c.display(); 27 } </pre>		<pre> Brand: Toyota Year: 2024 Car object destroyed === Code Execution Successful === </pre>

5.write a c++ program to create a class for a bank account with a constructor and destructor

Code:

```
#include <iostream>

using namespace std;

class BankAccount {
    string name;
    float balance;
public:
    BankAccount(string n, float b) {
        name = n;
        balance = b;
    }
    void display() {
        cout << "Name: " << name << endl;
        cout << "Balance: " << balance << endl;
    }
    ~BankAccount() {
        cout << "Bank account destroyed";
    }
};

int main() {
    BankAccount a("Ravi", 5000);
    a.display();
}
```

Output:

main.cpp	Run	Output
<pre>1 #include <iostream> 2 using namespace std; 3 4 class BankAccount { 5 string name; 6 float balance; 7 8 public: 9 BankAccount(string n, float b) { 10 name = n; 11 balance = b; 12 } 13 14 void display() { 15 cout << "Name: " << name << endl; 16 cout << "Balance: " << balance << endl; 17 } 18 19 ~BankAccount() { 20 cout << "Bank account destroyed"; 21 } 22 }; 23 24 int main() { 25 BankAccount a("Ravi", 5000); 26 a.display(); 27 }</pre>		<pre>Name: Ravi Balance: 5000 Bank account destroyed === Code Execution Successful ===</pre>